# Industry Applications of AI

*AI stopped being a research topic five years ago. Today every major Indian bank, retailer, hospital, and factory has a production AI system — and a hiring plan to build the next.*

L2

WHAT'S INSIDE

WHY THIS LESSON MATTERS

# AI is no longer a future technology. It's a current line item.

ROUTINE

### From R&D project to operational tool

Banks score loans with ML. Hospitals triage scans with CNNs. Telcos pick cell-tower upgrades with reinforcement learning. None of this is experimental any more.

LOCAL

### India is a top-3 adoption market

₹4 lakh crore Indian AI services market by 2027 (NASSCOM). UPI, Aadhaar, and ONDC produced data layers that AI naturally sits on top of.

HIRING

### Every domain is hiring AI talent

FinTech, HealthTech, AgriTech, GovTech, EdTech, Defence — each is at a different maturity stage. Knowing which fits your background changes your career.

YOURS

### Pick where you actually want to work

Choosing the domain is half the job. Same AI skills, vastly different impact when applied to disease screening vs. ad click-through prediction.

LEARNING OUTCOMES

# By the end of this lesson, you will be able to —

1. Name three AI capability patterns: perception, prediction, generation.
2. Map at least two production AI systems to each of six industries.
3. Explain three reasons India's AI market is structurally different.
4. Match your existing degree/experience to a likely AI domain.
5. Read an AI press release and judge: hype, pilot, or real adoption.

AGENDA

# What we'll cover today

*Four short parts, ~12 minutes each. Each builds on the last.*

**01**

## AI capability map

Three patterns: perception, prediction, generation. Almost every product fits one.

**02**

## Industry deep-dives

Six sectors, two production examples each — Finance, Healthcare, Retail, Manufacturing, Agri, Cyber.

**03**

## India context

What our market does differently. Aadhaar, UPI, ONDC, regional language gap.

**04**

## Picking your fit

Map your background to a domain. Read announcements past the marketing.

PART 1 OF 4

01

# AI capability map

*Three patterns explain almost every product in this lesson.*

CONCEPT 1 OF 9

# Pattern 1 — Perception. Convert raw signal into structured data.

Perception = take pixels, audio, or text and produce a label or extracted record.

Computer vision in radiology: chest X-ray → flag pneumonia probability. Apollo, AIIMS, Tata 1mg all run versions of this.

Speech-to-text on call centres: 100,000 customer-support calls / day at a major telco → automatic transcripts → topic clustering for QA.

Document AI in banks: scanned KYC documents → extracted name + DOB + address fields, no human keying. SBI, HDFC do this at scale.

Common technology stack: CNNs (vision), Whisper (speech), LayoutLM (documents).

KEY POINTS

- Input: pixels, audio waveform, or scanned document.
- Output: a label, transcript, or extracted record.
- Usually replaces a tedious human review step.
- Bottleneck = labelled training data, not models.

CONCEPT 2 OF 9

# Pattern 2 — Prediction. Score, forecast, recommend.

Prediction = take structured features and produce a number or a ranking.
Credit scoring at fintech NBFCs (KreditBee, ZestMoney): 50+ features → default probability → loan decision.
Demand forecasting at Flipkart, Amazon India: SKU × pincode × day → expected order count → pre-position inventory.
Click-through prediction in ad auctions: user × ad × context → CTR → bid price. The model behind every ad you've ever seen.
Common technology stack: gradient boosting (XGBoost / LightGBM) still dominates tabular; neural nets win when there's lots of unstructured input.

KEY POINTS

- Input: structured rows (a database query result).
- Output: a score, rank, or forecast number.
- Easy to deploy; hard to keep honest as data drifts.
- First production AI for most teams.

CONCEPT 3 OF 9

# Pattern 3 — Generation. Create new content matching a pattern.

Generation = take a prompt and produce text, image, audio, or code that didn't exist before.
Chatbots and copilots: customer-support summaries, code completion (GitHub Copilot), marketing copy. Every IT services firm has at least 3 internal copilots.
Synthetic data: generate enough training samples to bootstrap a model in a domain where real data is scarce (rare diseases, fraud patterns).
Personalisation: regional language content generation — a single article auto-translated to 11 Indian languages with cultural localisation.
Common technology stack: LLMs (Claude, GPT, Llama, Mistral), diffusion models for images, vector databases for retrieval-augmented generation.

KEY POINTS

- Input: a prompt + retrieved context.
- Output: novel text, image, audio, or code.
- Newest of the three; most hype, biggest production risk.
- Evaluation is hard — "is this good?" isn't measurable like accuracy.

PART 2 OF 4

02

# Industry deep-dives

*Two real production examples per sector. Six sectors, twelve examples.*

CONCEPT 4 OF 9

# Finance & Healthcare — the two highest-spend sectors

FINANCE — Banks, NBFCs, insurers, payment companies.
HDFC / ICICI: ML-driven fraud detection on UPI transactions in < 200 ms. False positives are the real metric — too many = blocked customers.
Bajaj / Tata AIA insurance: claim triage models route fast-track claims for auto-approval, complex ones to human review.

HEALTHCARE — Hospitals, diagnostic labs, pharma, telemedicine.
Apollo / SRL Diagnostics: AI-assisted radiology — chest X-ray triage, cancer screening, retinal images for diabetic retinopathy.
Niramai (Bengaluru): thermal imaging + ML for breast-cancer screening, deployed across 40+ Indian cities.

KEY POINTS

- Finance has the most data, deepest ML maturity.
- Healthcare has the highest impact per model.
- Both regulated — explainability + audit trails non-optional.
- Highest-paid AI roles tend to cluster here.

CONCEPT 5 OF 9

# Retail · Manufacturing · Agri · Cyber — the breadth

RETAIL — Flipkart, Amazon India, Reliance JioMart, BigBasket.
Demand forecasting, dynamic pricing, search ranking, recommendation systems. Every product placement you see is model-driven.

MANUFACTURING — Tata Steel, Reliance refineries, automotive plants.
Predictive maintenance — vibration sensors on motors → ML model → flag a bearing 2 weeks before failure. Saves crores in unplanned downtime.

AGRITECH — DeHaat, AgroStar, ITC e-Choupal.
Satellite + weather + soil data → recommended crop, sowing date, irrigation schedule. Bridges expert agronomy advice to 30M+ farmers.

CYBERSECURITY — Quick Heal, Seqrite, in-house SOCs at every bank.
Anomaly detection on network traffic, phishing-email classification, automated incident triage.

KEY POINTS

- Retail = highest-volume models (millions of requests/min).
- Manufacturing = highest-margin models (one good prediction saves crores).
- Agri = highest-impact-per-rupee (advice changes farmer income).
- Cyber = adversarial; the model is racing the attacker.

PART 3 OF 4

03

# What's different about India

*Aadhaar, UPI, ONDC, and 22 official languages. The map looks different here.*

CONCEPT 6 OF 9

# India's structural data advantage — Aadhaar, UPI, ONDC

Aadhaar is the world's largest biometric identity system — 1.4B people, used for KYC across every regulated industry. AI systems can verify identity in a way no other country can.

UPI processes 18B+ transactions monthly — granular, real-time, low-cost transaction data. Credit scoring without a Bureau record, fraud detection in milliseconds, demand-side analysis at city level.

ONDC (Open Network for Digital Commerce) standardises commerce data across sellers — a data layer no other country has at this scale. Coming wave: cross-seller recommendation, supply-chain optimisation.

Cumulative effect: India trains models on data formats that don't exist elsewhere. International AI tools often need re-tuning to work here — and that re-tuning is the local moat.

KEY POINTS

- Aadhaar → identity AI is uniquely possible here.
- UPI → real-time transaction analytics at population scale.
- ONDC → commerce graph that's structurally different.
- International models often miss Indian context.

CONCEPT 7 OF 9

# The regional language gap — 22 languages, 800M+ speakers

Most AI advances are English-first. Indian language NLP lags English by 2-4 years on every benchmark — but the market it serves is far larger.

Bhashini (govt initiative) is open-sourcing high-quality Indic-language models — Hindi, Tamil, Bengali, Telugu, Marathi, Kannada, Gujarati, Malayalam.

Production use cases: customer-support chatbots in regional languages, voice-first interfaces for non-English-speaking users, content moderation across 11+ scripts.

If your career goal is impact-per-user, the regional-language AI opportunity is the largest under-served segment in the world right now.

The teams hiring fastest here: AI4Bharat, Karya, Reverie, Translingua, plus internal NLP teams at every Indian IT services major.

KEY POINTS

- Indian-language NLP is 2-4 years behind English.
- The market is 800M+ speakers; growing share goes online yearly.
- Bhashini-backed open models are leveling the field.
- Underserved → high impact per hire.

PART 4 OF 4

04

# Picking your fit

*Map your background to a domain. Read announcements past the marketing.*

CONCEPT 8 OF 9

# Match your background to an AI domain

COMMERCE / ECON / FINANCE BACKGROUND → FinTech credit scoring, algorithmic trading, insurance claim prediction. Your edge: you read balance sheets, not just code.

BIOLOGY / MEDICINE BACKGROUND → HealthTech screening, drug discovery, hospital operations. Your edge: clinical context the engineer doesn't have.

AGRICULTURE / GEOGRAPHY BACKGROUND → AgriTech, climate AI, GIS. Your edge: domain reality that satellites alone can't teach.

MECHANICAL / INDUSTRIAL ENGINEERING → Predictive maintenance, robotics, process optimisation. Your edge: knowing why a machine actually breaks.

ARTS / LANGUAGES → NLP, content moderation, localisation, AI policy. Your edge: judgement on what a model output means in context.

PURE CS / IT → Anywhere; pair with one of the above by reading the domain literature.

KEY POINTS

- Your degree is a head-start, not a ceiling.
- Domain knowledge + AI skill beats pure AI in production.
- Pick a domain you'd argue about for free.
- First job is leverage; second is impact.

CONCEPT 9 OF 9

# How to read AI announcements like an analyst

Every week a company announces an "AI strategy". Most are press releases. A few are signal.

Three questions separate them:

1. Is there a NAMED PRODUCTION USER? If "we're piloting with one bank" → pilot. If "5M users across 200 branches" → real.
2. Is there a REVENUE LINE? Real AI shows up in the P&L. "AI Cloud revenue grew 38%" is a fact; "AI will transform our business" is a wish.
3. Can the company HIRE TO IT? Job postings are cheap to write. Look for sustained 6-month hiring in specific AI roles — that's where the money actually went.

When all three are present, the company is operating. When none are, it's a press release.

When 1-2 are present, it's a serious bet but unproven.

Same lens on your own choices: pilot, real, or wish?

KEY POINTS

- Named users > vague claims.
- Revenue line > strategy slide.
- Sustained hiring > one role.
- Apply the lens to your own decisions.

PAUSE & REFLECT

# Where do you naturally fit?

BACKGROUND

## What's your degree / domain?

Engineering, science, commerce, arts — every one has an AI path. Most students underestimate non-engineering fits.

PROBLEM

## What problem bothers you?

AI gets boring when you don't care about the domain. Bored engineers ship bad models. Pick a problem you'd debate over chai.

CONSTRAINT

## Where can you actually get hired?

Pune, Bengaluru, Hyderabad, NCR have most AI roles today. Tier-2 is growing fast. Remote AI work exists but is competitive.

KEY TAKEAWAYS

# Five things to remember

1. Three AI patterns explain ~90% of production systems: perception (see/hear), prediction (forecast/score), generation (create).
2. Every major Indian industry has AI in production now — Finance leads, Healthcare and Agri are the fastest-growing.
3. India's AI advantage is data — Aadhaar identity, UPI transactions, ONDC commerce — none of which other markets have at this scale.
4. Your degree is not your destiny. Commerce + ML = FinTech credit scoring. Biology + ML = HealthTech screening. Pick the overlap.
5. Press release vs reality: ask if there's a production user count, a revenue line, or a customer testimonial. If not, it's a pilot.

LESSON 3 — PYTHON PROGRAMMING ESSENTIALS

# From the map to the tools.

*You now know where AI gets used. Next you'll start building it. Python is the working language of every team you've heard about today — Tata, Reliance, Flipkart, Paytm, every healthcare startup. Variables, control flow, functions, and the muscle memory you need for the next 11 weeks.*

- Variables, types, control flow
- Functions and modules
- Reading the Python docs
- Your first end-to-end script